The Foundation, Explained: $(\mathbb{S}, \tau, \omega)$ and Unsharp Unity

Harald Kjøge Rønning

Companion to Foundation, Revision 24, and to Becoming, Revision 12. Every technical term built from the ground up — written for a reader with no prior background in this specific mathematics, not for a specialist.

Revision 11.

---

1. The triple

1.1 What $\mathbb{S}$ is, and how it is built

Start with the ordinary real numbers. There is a well-known trick, older than a hundred years, for "doubling" a number system into a bigger one: take *pairs* of numbers from the old system, and define a new multiplication rule that combines them. The oldest example is how the complex numbers are built from the real numbers. A complex number is a pair of real numbers (a, b), usually written $a + bi$, with

$$(a, b)(c, d) = (ac - bd,\ ad + bc).$$

Nothing mysterious happens except a new symbol i appears with $i^2 = -1$; multiplication stays commutative and associative.

Apply the same trick again — pairs of complex numbers — and you get the **quaternions**, four-dimensional, discovered by Hamilton in 1843. Something is lost: quaternion multiplication is no longer commutative.

Double again — pairs of quaternions — and you get the **octonions**, eight-dimensional. Something further is lost: octonion multiplication is no longer even associative.

Double once more — pairs of octonions — and you get $\mathbb{S}$, the **sedenions**, sixteen-dimensional. What is lost this time is the property of being a division algebra: two nonzero sedenions can multiply to give exactly zero, something impossible for real or complex numbers. This staircase — $\mathbb{R} \to \mathbb{C} \to \mathbb{H} \to \mathbb{O} \to \mathbb{S}$ — is the Cayley–Dickson construction, and $\mathbb{S}$ is where this project's formalism begins.

Concretely: $\mathbb{S}$ has sixteen basis directions, labeled by the sixteen four-digit binary strings 0000 through 1111. Multiplying two basis directions gives, up to a sign, the direction labeled by combining the two labels bit-by-bit. The whole multiplication table reduces to just one thing: which of these products come out plus, and which come out minus — and that pattern is completely fixed by the doubling construction above, not chosen freely.

A few properties, stated with equal weight to their failures, since a foundation should own its liabilities as plainly as its virtues:

- **Unital**: there is a direction that leaves everything unchanged when multiplied.
- **Power-associative**: raising a single element to a power is unambiguous.
- **Positive-definite norm**: every nonzero element has a strictly positive "length."
- **Not associative. Not alternative. Has zero divisors** (two nonzero elements whose product is zero) — real features of $\mathbb{S}$, not omissions from a list of virtues.

1.2 The symmetry group

Aut($\mathbb{S}$), the automorphism group, is every way of relabeling or rotating the sixteen directions that leaves the multiplication table unchanged — exactly as a cube has a symmetry group of rotations that leave the cube looking the same. For $\mathbb{S}$ this splits into two pieces: a large, continuous piece G_2 (fourteen independent directions of rotation), and a small, discrete piece with six elements, identical in structure to S_3, the group of ways to shuffle three labeled objects. What that shuffling group actually shuffles is the subject of the next section.

1.3 The split τ, and a free choice hiding inside it

$\mathbb{S}$ contains, hidden inside it, three different but entirely equivalent eight-dimensional copies of the octonions $\mathbb{O}$ — one step back in the doubling staircase. (These are not the only eight-dimensional structures $\mathbb{S}$ contains, but they are the three singled out by this specific doubling relationship, and the only ones this construction needs.) The six-element shuffling group from §1.2 is exactly what permutes these three choices among each other. This is an exact, algebra-level fact, with nothing tentative about it: the three copies sit cleanly inside $\mathbb{S}$ itself, genuinely conjugate under honest automorphisms of that one shared object. A separate, harder subtlety arises only later (§2.12), once each copy gets used to build an actual quantum construction from it — the three resulting quantum structures are, at that point, separate constructions in their own right, and whether they can be identified with one another is a different question from whether the underlying algebraic copies relate cleanly, which they do.

τ is the choice of *one* such copy: a "flip" with $\tau^2 = 1$, leaving one eight-dimensional half entirely alone (call it $\mathbb{O} = \mathrm{Fix}(\tau)$) and reversing the sign of the complementary half ($\mathbb{O}e_8$). None of the three possible choices of τ is privileged; they are related by a further symmetry μ, of order three (apply it three times, return to the start). Which one gets used is a free, symmetric choice — an *orbit coordinate*, in the sense that moving along the orbit of equivalent choices changes nothing physical.

A second free choice hides inside $\mathbb{O}$'s own seven-dimensional imaginary part, and it matters later (§2.1) because a specific direction there gets singled out to do real work: G_2 acts **transitively** on the sphere of unit-length directions inside this seven-dimensional slice — meaning any one such direction can be rotated into any other by some symmetry in G_2. Practically: whichever direction later gets used as a special reference point (called the *mediator*), the choice of *which* direction is free, the same way choosing which point on a perfectly round sphere to call "the top" is free — nothing about the sphere itself prefers one point over another.

1.4 Where the dynamics comes from — not a further free choice

Physical theories usually need a notion of "flow," of change over time, supplied as extra data on top of whatever static structure has already been built. Here, remarkably, no such extra data is needed.

There is a standard mathematical technique called a *contraction*: take a curved or twisted structure and controllably "flatten" it, the way slowly flattening a globe into a map preserves some features and loses others — and, sometimes, reveals a symmetry invisible on the original curved version. $\mathbb{O}$'s seven-dimensional imaginary slice undergoes exactly this kind of contraction in this project's prior work, and something specific happens: a *dilation* symmetry appears on the flattened structure that was not present on the original curved one — not one single, uniform stretch, but a whole *family* of rescalings, one for each way of stretching two specially paired directions (playing the roles position and momentum play in ordinary physics) by different amounts. This family survives every later stage of the construction as an exact symmetry, and it has one uniquely singled-out one-parameter subfamily: the rescalings that stretch one of the paired directions by exactly the factor they shrink its partner, leaving their product — and with it, a built-in constant playing the role $\hbar$ plays in ordinary physics — completely unchanged.

This is, precisely, what *squeezing* means (§2.6 explains the full physical picture): stretching one direction of quantum uncertainty while compressing its conjugate partner, keeping the total fixed. That squeezing transformation is the **squeeze flow** used throughout the rest of this document — the thing that supplies "time" and "change" to everything built from here on.

The important point: this flow is *derived*, not chosen. Given $(\mathbb{S}, \tau)$ and the mediator's free choice already covered, nothing further needs to be specified to get a working notion of dynamics.

Honest limit, stated plainly: the flow's *existence* rests on a specific contraction scheme — its precise grading and scaling pattern. That scheme is forced once you fix an assumption first: that there are three "coordinate" directions and three "momentum" directions to begin with. Given that starting shape, which specific assignment works is pinned down completely, checked exhaustively over every possibility. *Why* three of each — rather than some other split — is not derived anywhere in the cited work, and is now a named open question in its own right, not a hidden assumption. This is the single most load-bearing citation in the whole document, and it has now been checked end-to-end within this project, in a standalone companion document: this rescoping is the result of that check, not an unverified caveat awaiting one. One further piece of that check is worth having in plain language: one alternative shape actually got tried, not just imagined — with a different, four-level pattern instead of three, the clean split between "position-type" and "momentum-type" directions dissolves entirely, neither side keeping a clean, exclusive claim to being the special one. So the three-ness of space and the existence of a position/momentum pairing at all turn out to be the same open question wearing two faces, not two separate mysteries — and whether *any* other starting shape can be made to work is a finite list to check, not an infinite one, though nobody has run that check yet.

1.5 The state ω

An algebra like $\mathbb{S}$, once extended to its full operational form, is best thought of as a grammar — the complete list of everything that could, in principle, be measured, and the rules relating those measurements to each other. It does not, by itself, say what any particular physical situation is actually doing. That further fact is a **state**: a rule assigning an honest expectation value to every measurable quantity, consistent with the grammar's own rules.

A state does more than sit there abstractly, though. There is a standard recipe (named for Gelfand, Naimark, and Segal — "GNS") for turning a state into an actual mathematical *world*: a space of quantum vectors the algebra's operators genuinely act on. Two different states can generate the *same* world in disguise (technically: *quasi-equivalent*) or two genuinely *different*, walled-off worlds (technically: *disjoint*) that no ordinary, *reversible* physical process could ever bridge — a qualification that matters later: the framework's own hoped-for mechanism for genuine change (§2.11) is explicitly not of this ordinary, reversible kind, which is exactly why it is able to attempt a job disjointness otherwise forbids. The collection of every state reachable from a given one by ordinary reversible means — everything quasi-equivalent to it — is called that state's **folium**: its local neighborhood of possibility, as distinct from some entirely separate universe.

ω is a *regular quasi-free state*, generating its own such world of type III_1 (§2.2 explains what "type" means) — but ω is deliberately **not** one uniquely singled-out state. It stands for a generic member of a whole family of such states, whichever one actually turns out to be physically realized. Which member that is counts as physical data, empirical input — not something the formal starting point specifies.

The construction of the operational algebra itself, and the results this document leans on, come from a specific, named prior body of work — five papers in total, with partial locators for two of them and one more specific pointer besides. A paper on the second contraction (the flow above, and the symmetry derived in §2.1); a paper on the squeeze-thermal state (the type verdict, and — a specific, named result within it — the proof behind §2.1's rotation argument); a paper closing a last construction-specific gap; a paper proving state uniqueness at the strongest available level, the fact §2.1's proof leans on; and a paper establishing that temperature is this construction's only genuine parameter, the fact §2.9 leans on. None of it is re-derived here; full citation locators remain owed for most of it — a debt this document owns rather than hides. **A sixth document now exists:** a standalone compilation of the whole chain from these five into one place, checkable start to finish. It confirms most of the chain outright, closes the question of a technical property the uniqueness theorem needs — the precise property Haagerup's own theorem is stated for, by a recorded argument rather than a bare assertion — and — its most important discovery — locates a genuine gap that had been hiding inside the "single most load-bearing citation" above: the grading choice §1.4 depends on is forced only once you *already* assume three coordinate directions and three momentum directions; that starting assumption itself is not derived anywhere, and is now this project's most interesting open question rather than a hidden premise. And a further honesty note: the checking this project has done on those results, within its own collaborative process, is real, but it is not the same thing as independent peer review or formal proof-checking.

1.6 The emergence axiom

The order in which a formal theory is *built* — algebra first, state second — is not a claim about which one is more fundamentally real. $\mathbb{S}$ is treated as the best formal description available of something that may, in itself, lie outside any algebra, metric, or vocabulary whatsoever; ω is that ground's sole imprint on the formalism — not one formally singled-out instance of the ground, but the one *kind* of object (a state) through which the ground shows up at all. Which specific member of the family is realized is, again, empirical input, not formal data.

$(\mathbb{S}, \tau, \omega)$ is the whole of the formal beginning. Nothing further is added.

---

2. Unsharp unity

2.1 A further symmetry, derived in two exact steps

Fixing the mediator direction from §1.3 breaks G_2 down. The break happens in two precise stages, each checked two different ways — a general mathematical argument, which is what actually proves it, and a separate numerical spot-check on a random example, confirming the argument to extraordinary precision without substituting for it.

Stage one. Fixing the mediator alone leaves exactly the symmetries that also fix that one direction — and by a classical fact about G_2 acting on that seven-dimensional sphere, this collection forms a specific eight-dimensional group called $\mathfrak{su}(3)$.

Stage two. Also insisting that a further, related structure — three particular directions, paired with their partners by the mediator's own multiplication — stay *real*, rather than mixing with their complex-structure images, cuts this down further, to the ordinary three-dimensional rotation group $\mathfrak{so}(3)$: the everyday symmetry of turning an object in space.

This same rotation symmetry survives every later stage of the contraction described in §1.4, remaining an exact symmetry throughout. Separately — and this needs its own extra step, not automatic — it also leaves the *specific state* ω unchanged, rather than merely shuffling equally-good alternative states among each other, and this rests on a proof, not merely a reasonable expectation: rotations act by a specific kind of structure-preserving transformation that keeps a state's overall character intact, and are proven — not just checked numerically, though the numerical check is there too — to commute with the derived flow. Given this, rotating ω produces another equilibrium state of that very same flow at that very same temperature; and since the relevant uniqueness theorem holds at its strongest available level — covering every state of the general kind involved, not some narrower class — that rotated state must be ω itself, with no further gap to fill.

2.2 What "type" means for an algebra of measurements

Mathematicians classify the "universe" of measurements an algebra generates into different **types**, based on one deep structural question: is there a sensible, evenly-weighted way of totaling everything in it — a generalization of the ordinary trace of a matrix, called simply a *trace*?

- **Type I** universes have a good notion of trace and are built from familiar, discrete building blocks — ordinary textbook quantum mechanics.
- **Type II** universes are stranger — infinitely many possibilities, but still a sensible, if subtler, notion of "how much."
- **Type III** universes have no trace *at all*, not even a partial one — nothing in them can be evenly weighed or totaled.

What forces Type III is not simply a matter of *how many* possibilities are involved — a system can have infinitely many possibilities and still be Type I, as an ordinary, if very large, quantum system is. What forces Type III is the *kind* of state and the *kind* of region: genuine thermal-equilibrium states of infinite systems, and local patches of a quantum field (as opposed to the whole field examined at once), generate Type III worlds. Not sheer size, and not interaction — this specific combination. It is exactly the situation here, because the construction is genuinely thermal and infinite in the relevant sense.

Within Type III there is a finer classification still (called III_0, III_λ, III_1), based on how rich the internal flow of time a state generates turns out to be. This project's operational algebra is established, elsewhere and cited rather than re-derived here, to be of the richest kind: **Type III**$_1$ specifically.

2.3 No trace, ever — one end of unsharp unity closed

A state that is completely static — flowing nowhere, changing in no way — turns out, as a matter of general mathematical fact, to be *exactly* the same thing as a trace: flow trivial if and only if tracial. Since nothing in this operational algebra is ever a trace (established for every state within a world of the established type), nothing is ever completely, permanently settled either. Total settlement, in the sense of everything having settled so completely that no further change could ever occur, would require exactly this kind of state — and none exists.

2.4 Two worlds can be the same, or genuinely different — and nothing else

Recall from §1.5: a state can generate a world quasi-equivalent to another state's world (the same world, differently described) or a *disjoint* one (a genuinely separate universe, walled off by more than mere distance). For an important class of states called *factor states* — which everything discussed here belongs to — these two possibilities are, provably, the *only* two: no partial or in-between cases.

2.5 The symmetric reference

The reference configuration behind "pure unity" — the earlier, more informal language for the unity horizon called it "the free scaffold vacuum" — is the pure, unsqueezed, coldest-possible configuration: the ordinary quantum vacuum, in the most literal sense. Notice this is *not* a counterexample to §2.2's account of Type III: the pure vacuum is neither a thermal-equilibrium state nor a local patch — it is the global, coldest-possible configuration — which is exactly why it lands in Type I instead.

Its separation from any genuinely thermal configuration needs no delicate estimate at all. A *pure* state generates a Type I world (it is about as simple and sharply-defined as a quantum state can be); a genuinely thermal one generates a Type III world. Worlds of different type can *never* be the same world in disguise — this is a basic, general fact, nothing to do with distance or approximation, only with the *kind* of object each one is. So the reference is provably, structurally separate from any genuinely thermal configuration.

One precise qualification, honestly carried rather than smoothed over: this separation is *unconditional* relative to ω's own world, since that world's type is established directly. Relative to every *other* member of the family, it currently rests on the weaker, cheaper requirement that the other member's world merely fail to be Type I — not on the full, richer III_1 verdict, which is a separate, more expensive thing to establish for every member uniformly and remains open.

2.6 Squeezing, and the shape of quantum uncertainty

Heisenberg's uncertainty principle says a quantum system's "position" and "momentum" can never both be known perfectly at once — there is always some minimum, irreducible fuzziness. *Squeezing* is a real, well-studied physical operation — used, for instance, in the most precise gravitational-wave detectors — that trades fuzziness between the two directions: position can be made extra sharp at the cost of momentum becoming extra fuzzy, or the reverse, while the *total* amount of fuzziness, in a precise technical sense, stays exactly fixed. This is exactly the mathematics behind §1.4's derived flow. A **covariance** is simply the bookkeeping device tracking how much fuzziness sits in which direction, for a given state. The coldest, perfectly balanced, unsqueezed state — the vacuum — has the minimum possible fuzziness, spread evenly. Warming a system adds *more* fuzziness overall, on top of whatever squeezing has already been applied.

2.7 The two endpoints, in this vocabulary

The operational algebra here is built from *bosons* — force-carrier-type quantum objects — not the matter-type objects (fermions) discussed in §3. For bosonic quantum states of this kind, the covariance can never fall below a fixed minimum, and equality holds exactly at the pure, unsqueezed vacuum: $\frac{1}{2}$ marks the *symmetric* end, not the settled one — worth flagging, since the natural first guess (that a maximally mixed, "washed-out" state is the symmetric reference) runs backward. The completely settled, flowless end — the trace — is not a squeezed-or-warmed *version* of anything at all: it fails to be the ordinary kind of quantum-field state in the first place, lacking even the continuous field structure a genuine physical configuration needs. The two ends differ in *kind*, not merely in degree.

2.8 Thermal states, and why different temperatures are different worlds

A rigorously defined notion of "thermal equilibrium at a given temperature" exists for infinite quantum systems — called a KMS state, after the physicists Kubo, Martin, and Schwinger. A standard, well-established fact (cited to a classic textbook on the subject, not derived fresh here) says: for the *same* underlying dynamics, genuinely different temperatures *always* generate disjoint worlds — no exceptions, and no delicate estimate required, provided the dynamics is nontrivial and the states in question are not themselves traces (the one degenerate edge case, already excluded by §2.3's fact).

2.9 A family of worlds, not one world

Putting this together: the natural collection of thermal configurations this project studies is *not* a collection of different states within one world — apart from possible rare coincidences, it is a collection of genuinely separate worlds, one per configuration. "Admissible," accordingly, does not mean "belonging to one specific world." It means belonging to *any* of them — the whole union.

The family has exactly one knob, not two — not a temperature and a separate "squeeze amount," despite how earlier documents in this project once spoke of it. The technical sources this project cites state this outright: temperature "remains the one genuine physical parameter of the state," and a further theorem proves it at full strength — allowing for the *most general* possible version of such a state, including squeezed ones, the equilibrium condition itself forces any squeeze content to exactly zero. There is only one knob, and every section below is written accordingly.

2.10 χ, precisely — settling depth, defined

Given all this, χ — the informal "distance from pure unity" dial of earlier documents — now has a precise mathematical definition, and the word "distance" needs one correction before going further: χ is not a distance measured *within* any single world. The worlds it labels are disjoint, and disjoint worlds turn out to be maximally and identically far apart from each other in the ordinary technical senses of "far" — nothing about *how much* farther survives once two configurations are in different worlds at all. But χ is not a mere unordered label either. It is a genuine, ordered coordinate on the *space of worlds themselves* — closer, in spirit, to floors of a building than to either a ruler or a name tag. Floor 12 is genuinely, physically higher than floor 3; but there is no path from one floor to the other within either floor's own interior — reaching one from the other needs an elevator, and, per §2.11, this framework does not yet have one. Not a distance to be estimated, then, but a specific invariant extracted from a state's covariance: its **symplectic spectrum**. There is a standard trick — analogous to finding the natural principal axes of an ellipse — that pulls out of *any* covariance a small set of numbers immune to squeezing: only genuine warming changes them. For the systems here, there is one such number per mode, ranging from $\frac{1}{2}$ (coldest, perfectly balanced) up toward infinity (hottest, most washed-out) as temperature rises, and it is exactly *independent* of squeeze throughout.

For a state genuinely at thermal equilibrium — a true member of the family — one shared temperature coordinates every mode's number together, so χ reduces to that single figure. Off equilibrium, more care is needed: a generic state has no reason for its modes to share one parameter, so its collection of per-mode numbers may not reduce to one number at all, and comparing the "depth" of two such generic states can be genuinely ambiguous — closer to comparing two hands of cards than two heights. This is not a flaw specific to this framework: ordinary thermodynamics works the same way. Temperature is a single, well-defined number only for a system genuinely at equilibrium; off it, energy is simply spread however it happens to be spread, with no one number summarizing it. A dial that stops being a single number away from equilibrium is behaving exactly as a real physical dial should.

Within the family, the single-number picture holds cleanly and without qualification: every member reads exactly $\chi = \frac{1}{2}\coth(\beta k/2)$ at its own temperature (k the mode frequency, kept distinct from ω the state), with no second, independent knob to complicate the count.

One piece of unfinished business, honestly named: making this work properly across the entire, infinite-mode system — not just one mode in isolation — requires carefully handling modes whose natural frequency runs to zero, where the naive per-mode number blows up regardless of temperature. This has not yet been fully worked out, and it is the single largest remaining *mathematical* task tied to χ's own definition — a different kind of gap from the *conceptual* one named in §2.11 below.

2.11 χ never changes on its own — a genuine discovery

Because χ is built from exactly this squeeze-and-warming-independent invariant, and because the flow derived in §1.4 is exactly the kind of transformation that *preserves* this invariant by its own mathematical nature, it follows as a matter of pure logic — not a further assumption — that this flow can *never* change χ, for any state it acts on, ever.

This is, properly understood, the *correct* signature of a good "which phase am I in" label, not a flaw in the definition. But it carries a consequence worth stating plainly rather than glossing over: whatever mechanism actually drives *settling* — a genuine change in depth over the universe's history — cannot be this flow. It would have to be something of a fundamentally different kind: something that genuinely loses information, or interacts irreversibly with an environment, rather than the tidy, reversible transformation this flow represents. Nothing of that kind currently exists anywhere in this framework's toolbox. This is the single most conceptually important gap the whole framework currently carries — a different kind of priority from §4's ledger, which orders items by how much currently *depends* on each rather than by how large the gap itself is; the citation debt ranks first there because more rests on it, not because it is a bigger absence in its own right.

χ measures *depth* of settling specifically, and says nothing by itself about *which* split or mediator direction (§2.12) a settled configuration might select — that is a different kind of question, not a further blind spot of χ's own. At fixed temperature there is exactly one state, so there is nothing for χ to fail to distinguish there in the first place.

2.12 The remaining symmetry question

[This section draws on Becoming, a companion piece that has been through several rounds of review but remains younger and less settled than Foundation.]

Recall the free choices from §1.2–1.3: which mediator direction, which of the three equivalent splits. It remains genuinely open whether the *actual*, physically realized configuration treats this choice as forever free, or whether settling itself, over time, singles out one particular direction or split as realized — the way a compass needle, once an actual magnetic field is present, does pick out a specific direction even though "north" was arbitrary beforehand. If a version of the flow governing the choice of split turns out to act by the same squeeze-preserving mathematics as §2.11, and if that action is genuinely implemented on a core the three splits actually share, χ would be unable to distinguish which split or mediator gets chosen. But two things need checking before that conditional bites, not just one: whether the symmetry acts this way at all, and whether the three splits even share one mathematical space to act on in the first place — they are built from three different, only partially overlapping structures, and are at best equivalent in the abstract sense, not identical outright. Once those checks resolve, the live question would narrow to whether the choice happens *within* one level of settling depth, not whether depth itself is affected. None of this has been decided.

2.13 Which direction is "the past"

A second, separate claim: the near-symmetric end of χ's range is placed, specifically, toward the direction later identified as the past — not the future, not both alike. Unlike everything established so far in this section, this is a genuine assumption, not a proof: a zero-parameter boundary condition, defended by analogy to a respected proposal in ordinary cosmology (the idea, associated with Roger Penrose, that the very early universe's gravitational structure was exceptionally smooth, argued from symmetry rather than derived). It names a real choice between two provably distinct ends.

Because different worlds (§2.9) are, by definition, walled off from each other globally, any actual comparison of "before" and "after" — and with it, any genuine change in χ — could only ever be assessed *locally*, within a restricted patch of the whole structure, since globally disjoint worlds are typically still locally comparable in the ordinary way. This restriction to local patches is not a limitation added on top of the framework; given §2.11's finding, it is the *only* place any mechanism of settling could possibly live at all.

2.14 A declared limit — bosons only, for now

Everything in this section describes force-carrying (bosonic) matter specifically. If the matter-particle (fermionic) construction discussed in §3 eventually succeeds, it would bring in a different kind of quantum statistics entirely — one whose own "settled" and "symmetric" ends run *opposite* to the bosonic ones described here (their perfectly balanced, pure states sit at the *extremes* of possibility, and their fully-settled, trace-like state sits comfortably in the *middle* — the reverse geometry). If and when that happens, this whole picture of which end is which will need to be revisited for the combined system. Named honestly as a dependency now, not discovered as a surprise later.

---

3. The seed, stated as a commitment

3.1 The question

Modern attempts to build the particles of nature out of number systems like the octonions and sedenions rely on a mathematical trick, and that trick needs complex numbers — numbers involving i — to work. Should i be built into $\mathbb{S}$ itself, as an extra ingredient added by hand from the start — or can it arise instead from the state ω, with $\mathbb{S}$ remaining purely real?

3.2 Why complex numbers matter here, and where the real obstacle actually sits

The trick needs certain building blocks, called creation operators, to have an unusual property: applying the same operation twice in a row gives exactly zero. This is the precise mathematical origin of the Pauli exclusion principle — the rule that no two identical particles like electrons can occupy exactly the same quantum state. A building block with this self-cancelling property is what a *fermion*, the general category including electrons and quarks, is built from.

The simplest illustration is a single real number: a nonzero real number's square is always strictly positive, never zero, while $i^2 = -1$ supplies exactly the sign-flip needed. But this is only the apparent obstacle, offered as intuition — it is not, by itself, a general fact about every real mathematical structure (ordinary real matrices, for instance, can square to zero without being zero themselves). The *actual* obstacle proven here is narrower and more specific: it concerns elements of $\mathbb{S}$ itself, under $\mathbb{S}$'s own particular, positive-definite notion of length.

3.3 What was proven about real $\mathbb{S}$ — and what it actually rules out

No nonzero real element of $\mathbb{S}$ can have this self-cancelling property by serving *directly* as the required product — shown two independent ways: every nonzero real element has strictly positive length, and squaring it can never make that length vanish; and a separate, more careful argument, using a bookkeeping tool that sums an operation's effect across every direction at once, closes off a subtler potential loophole too.

This closes one specific route. It is *not* the obstacle a real fermionic construction actually faces. The standard recipe physicists already use for exactly this purpose — the same one used to properly describe Majorana fermions in ordinary physics — needs only an ordinary notion of length on a real space, which $\mathbb{S}$'s own positive-definite length supplies unconditionally, with no theorem required, the moment the needed structure is built abstractly from that length rather than borrowed from $\mathbb{S}$'s own multiplication.

3.4 Where the needed structure can come from instead

The real open question is therefore not whether $\mathbb{S}$ can supply the needed complexity at all — it can, trivially, via the standard route above — but *where the choice comes from* that turns that abstract structure into an actual, physical set of particle-creation operators. This choice (called a *polarization*) is a feature of which particular representation gets used, not of the underlying real structure itself. Several live candidates exist for supplying it: a dynamics selected by the state (this document's working choice); a seed complexified from the very start; the zero-divisor structure of $\mathbb{S}$ itself (elements that multiply to zero, mentioned in §1.1); or simply declining to build particles at all and working with the real structure directly.

The working commitment: $\mathbb{S}$ stays real; the needed choice is expected to come from ω, not be glued onto $\mathbb{S}$ from outside. This is motivated by a mechanism already demonstrated working, in full, for the bosonic (force-carrying) sector described in §2: a real classical system, the derived flow of §1.4, and a standard construction (due to Kay and Wald) yielding a genuine complex structure with nothing added by hand. Once a state exists, a separate, well-known piece of mathematics (Tomita–Takesaki theory) supplies that state's own natural internal flow automatically — but this certifies nothing further by itself; whether that flow *agrees* with the dynamics chosen to build the structure in the first place is a separate condition (called the KMS property), checked individually for each construction, which is exactly what the bosonic sector's own proof supplies. The state-selected route remains the working choice on precedent from that success, not from necessity.

3.5 An unexpected fact worth stating outright

If the operational algebra lands, as intended, on the richest, most standard kind of Type III_1 world (§2.2), a remarkable uniqueness theorem (due to Haagerup) says that its bare mathematical identity carries *no memory whatsoever* of $\mathbb{S}$, τ, or any detail of how it was built — every such world, however constructed, is the identical mathematical object. (This uniqueness is specific to the most standard, "well-behaved" kind of such world; genuinely different, non-standard versions do exist and are not covered by it.) Everything distinctive about *this* particular construction accordingly lives not in the bare type of the resulting world, but in what gets carried alongside it: the specific state ω and the family it belongs to, the rotation symmetry derived in §2.1, and — prospectively, once the matter-particle construction of §3.4 actually succeeds — a further sub-structure that construction would generate from τ's fixed half. Not yet, though: the current, force-carrier-only construction uses only τ's fixed half to begin with, so there is nothing further to point to there right now — that additional structure is a possible result the matter-particle sector alone could supply, not something already in hand.

Given this, it is worth naming precisely where $\mathbb{S}$'s *actual multiplication* — not merely its symmetry group, not merely its length function, since every sixteen-dimensional space with a positive-definite length is mathematically identical and carries no sedenion-specific content on its own — does work nothing generic could replace. One confirmed case: multiplication by the mediator direction serves directly as the complex-rotation operation in the derivation of §2.1's rotation symmetry. This is one instance, not a claim that the full accounting is complete.

3.6 What would change this commitment

This commitment holds provided the needed choice can, in fact, be supplied by a state-selected dynamics along the lines just described — matched, in detail, against the published mathematical construction of fermion content this project is checking itself against. That matching is not complete. If it is eventually shown that no such state-selected choice can reproduce what is needed, the live alternatives named in §3.4 — building complexity into the seed from the outset, foremost among them — become the honest fallback.

---

4. What this document is

A statement of starting data and its status, nothing more. It proves nothing about gauge structure, particle masses, or cosmology, and claims nothing beyond what is explicitly marked as established. Its only obligation is that every sentence carry its correct weight — a proven fact, an established proposition, an honest postulate, a stated commitment, or a named conjecture — so that anything built afterward inherits no hidden assumption it was never told it was making.

A running list of open items, kept here rather than scattered across correspondence: why the grading shape §1.4's flow depends on — three coordinate directions, three momentum directions — is assumed rather than derived, now that the standalone compilation has shown the *choice within* that shape is fully forced but the shape itself is not (§1.5); the mechanism of settling itself, which §2.11 shows cannot be the derived flow, confirmed for an even more basic reason on inspection (every family member is individually stationary under that flow) and which nothing currently supplies; the mode-density computation needed to make χ well-defined uniformly across the full infinite system (§2.10); whether every member of the family shares the richest type verdict uniformly, split into a cheap requirement (already sufficient for §2.5) and a more expensive one (needed elsewhere) — expected to resolve uniformly, but worth stating precisely why a *non*-uniform answer would be a discovery rather than a defect: this classification measures how rich a world's internal sense of time is, so if it varied along the family, that richness would itself be changing with settling depth; whether the choice of split gets physically selected by settling (§2.12, itself provisional — see that section's note); the direction postulate itself (§2.13); the condition under which the seed commitment of §3 would need revising; the reopening of §2's whole endpoint picture if the fermionic construction of §3 succeeds; precise citation locators for the remaining companion papers; the exact theorem number, within a cited textbook, for one fact used in §2.8; a complete accounting of where $\mathbb{S}$'s own multiplication does work nothing generic could replace, beyond the one instance named in §3.5; and, newly added, a three-part check that only becomes relevant if the matter-particle construction of §3 succeeds: whether it actually creates a genuine further sub-structure at all (right now it would not — the current, force-carrier-only construction already uses the whole of what would generate it, leaving nothing extra behind); whether that sub-structure, if created, has the right technical property to be classified in the standard way (a question about how it sits within the larger structure geometrically, not about any entanglement in the underlying physical state, which is a common but mistaken intuition); and whether a specific averaging-like operation exists mapping the larger structure back onto it, a genuine further requirement, not automatic once the first two hold; and, sharpening the split/mediator question of §2.12 further, two small finite computations neither done here: how much the three splits' underlying spaces actually overlap, and — on whatever they share — whether the relevant symmetry commutes with the flow there or instead links three separate copies of it together. This list changes as items resolve or new ones surface; it is kept here so the document remains its own most honest record.

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.